

AI 모델 성능 개선 결과

전체 강우

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)

-
- learning_rate = 0.0001 / 0.0005
 - Loss_function = MSE / MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습에 사용 (결측 5개 + 검증 3개)

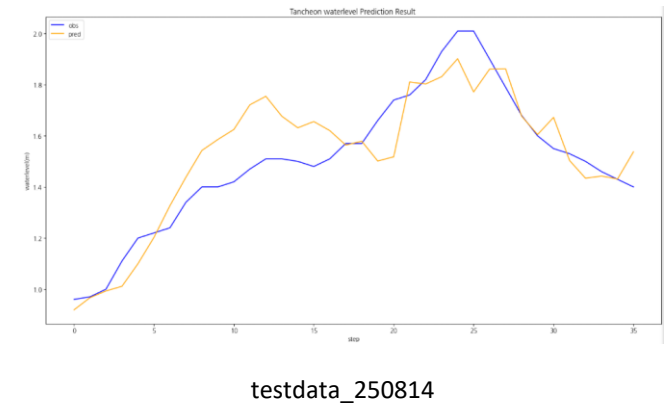
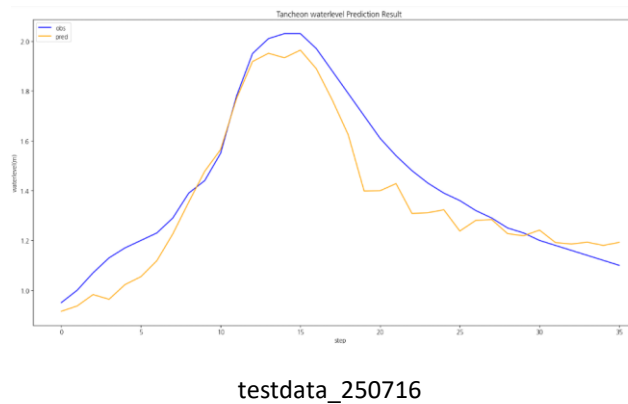
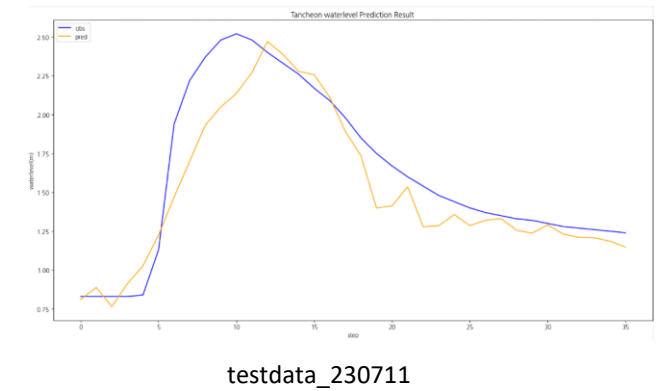
검증 : 학습 데이터에서 제외한 3개 + 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : ~ 2022 68번 (988개)
- 검증 : ~ 2024 96번 (282개)
- 테스트 : ~ 2025 109번 (142개)

num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - **궁내교**

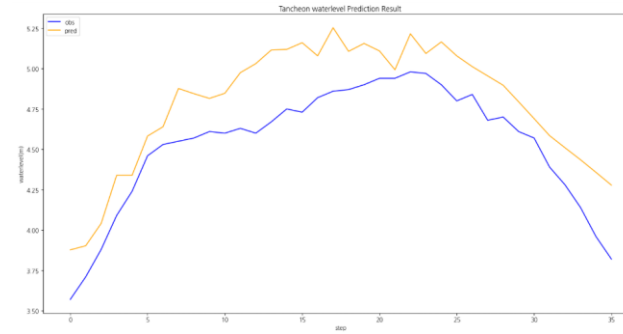
	궁내교
증설 전	testdata_220712 r2 : -0.2437 mse : 0.0493 mae : 0.1828 rmse : 0.2221
	testdata_230711 r2 : 0.8500 mse : 0.0419 mae : 0.1468 rmse : 0.2048
증설 후	testdata_250716 r2 : 0.8895 mse : 0.0110 mae : 0.0833 rmse : 0.1052
	testdata_250814 r2 : 0.7838 mse : 0.0152 mae : 0.0962 rmse : 0.1234



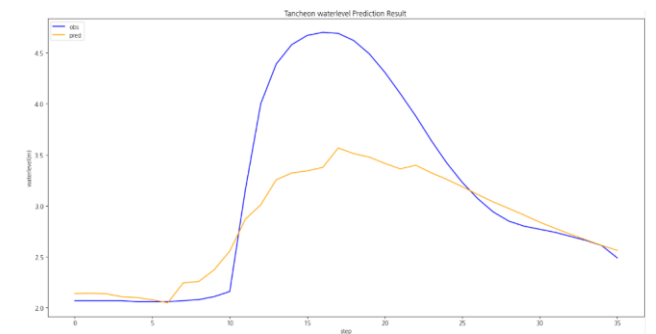
num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - 대곡교

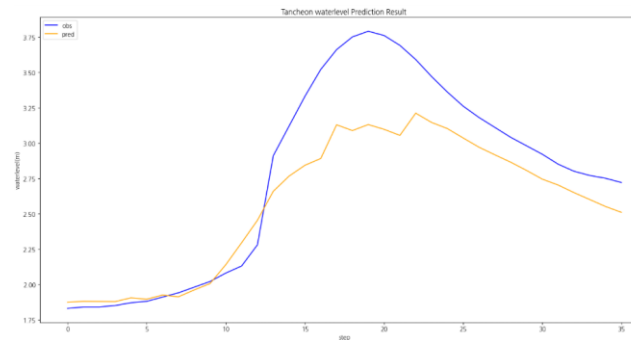
대곡교	
증설 전	testdata_220712 r2 : 0.4649 mse : 0.0749 mae : 0.2530 rmse : 0.2737
	testdata_230711 r2 : 0.6025 mse : 0.3617 mae : 0.3919 rmse : 0.6014
증설 후	testdata_250716 r2 : 0.7837 mse : 0.0998 mae : 0.2374 rmse : 0.3159
	testdata_250814 r2 : 0.6523 mse : 0.1206 mae : 0.2747 rmse : 0.3473



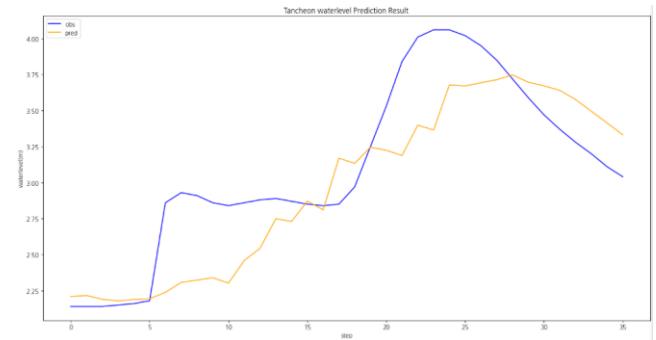
testdata_220712



testdata_230711



testdata_250716

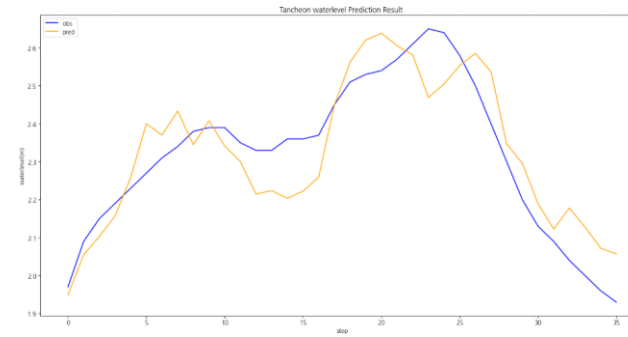


testdata_250814

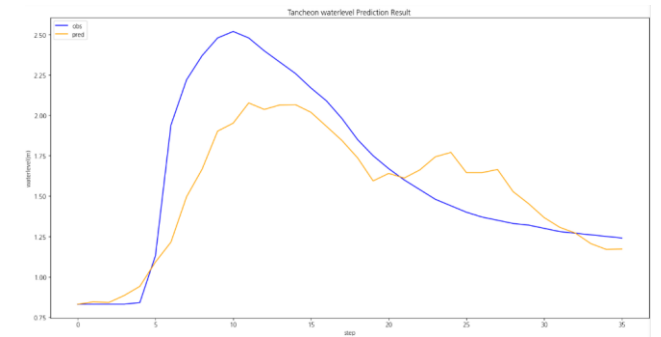
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - **궁내교**

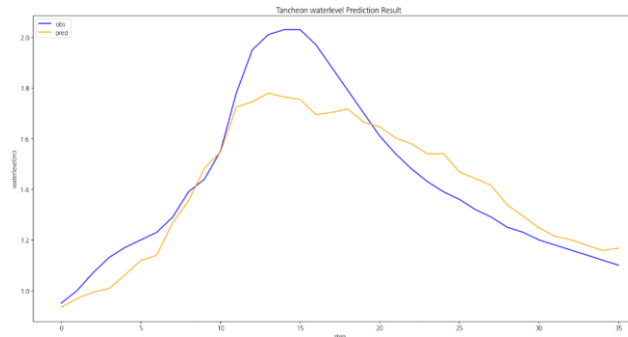
	궁내교
증설 전	testdata_220712 r2 : 0.7886 mse : 0.0083 mae : 0.07876 rmse : 0.0915 testdata_230711 r2 : 0.6794 mse : 0.0896 mae : 0.2132 rmse : 0.2994
증설 후	testdata_250716 r2 : 0.8537 mse : 0.0146 mae : 0.0960 rmse : 0.1211 testdata_250814 r2 : 0.7326 mse : 0.0188 mae : 0.0974 rmse : 0.1372



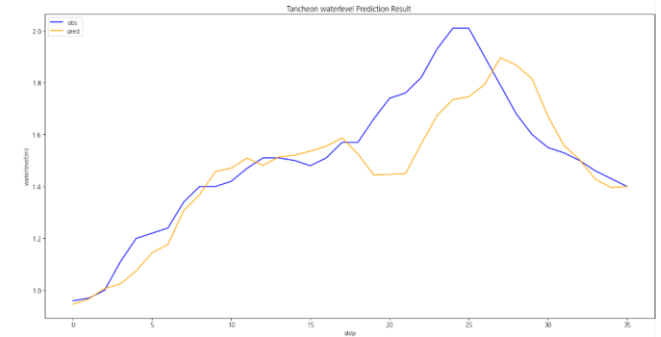
testdata_220712



testdata_230711



testdata_250716

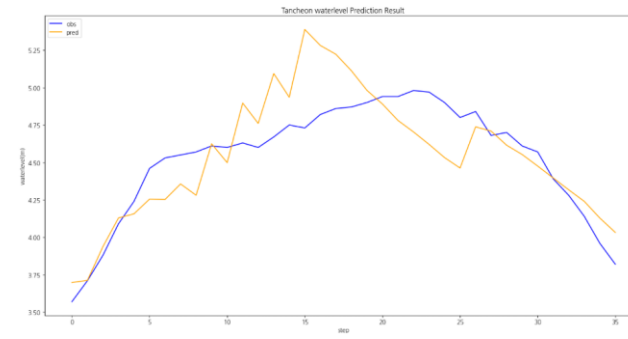


testdata_250814

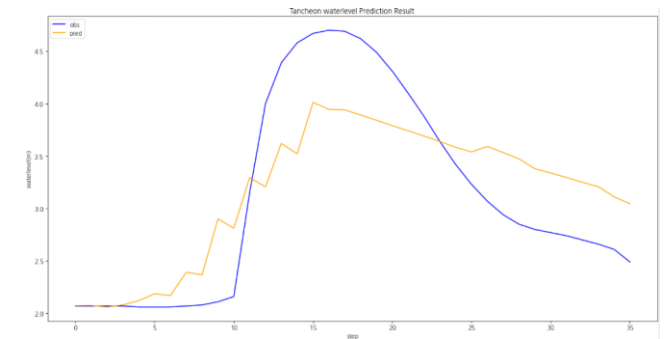
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - 대곡교

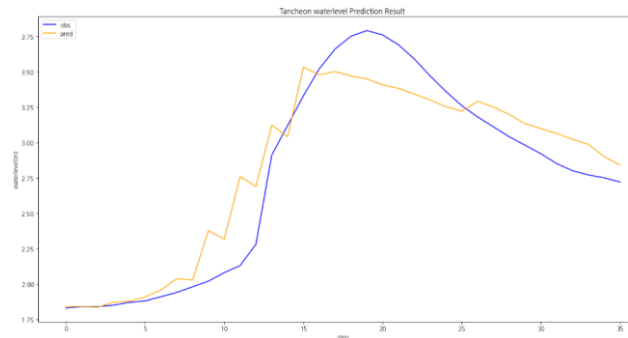
	대곡교
증설 전	testdata_220712 r2 : 0.5988 mse : 0.0561 mae : 0.1851 rmse : 0.2370
	testdata_230711 r2 : 0.6995 mse : 0.2734 mae : 0.4393 rmse : 0.5229
증설 후	testdata_250716 r2 : 0.8985 mse : 0.0468 mae : 0.1692 rmse : 0.2164
	testdata_250814 r2 : 0.6443 mse : 0.1234 mae : 0.2827 rmse : 0.3513



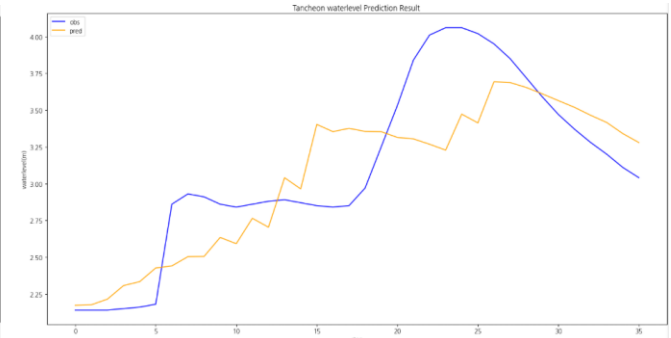
testdata_220712



testdata_230711



testdata_250716



testdata_250814

누적 강우 110mm 이상 4

학습데이터

같은 연도.다른 강우 패턴 대응 가능성 검증

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0001
 - Loss_function = MSE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

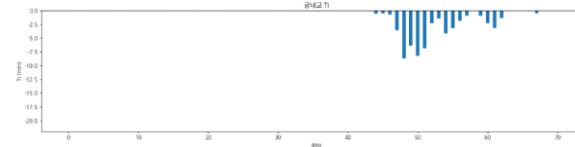
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

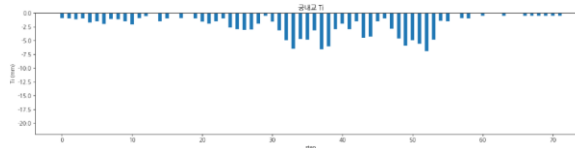
- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022(39) (1개)
- 테스트 : 2022(6667) (1개)

누적 강우 110mm 이상 4 예측 코드 - **궁내교 1**

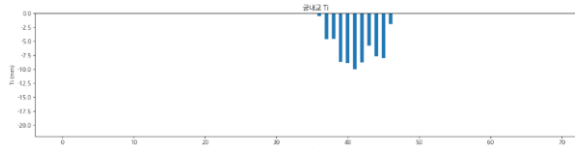
	궁내교
testdata_200802	r2 : 0.7237 mse : 0.0584 mae : 0.1572 rmse : 0.2417
testdata_220712	r2 : 0.3192 mse : 0.0270 mae : 0.1369 rmse : 0.1643
testdata_230711	r2 : 0.8533 mse : 0.0410 mae : 0.1509 rmse : 0.2025



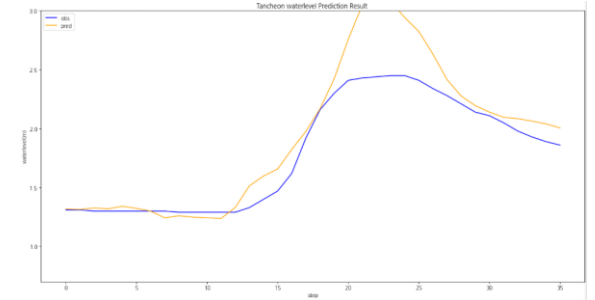
testdata_200802



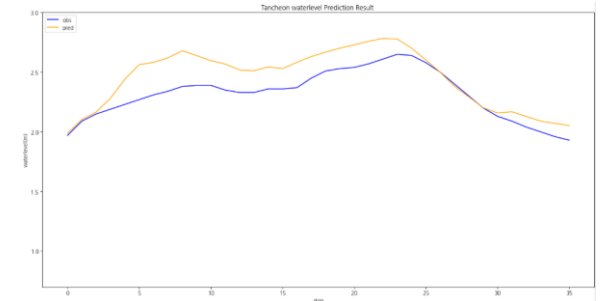
testdata_220712



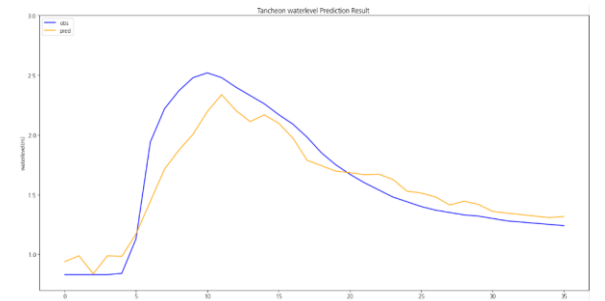
testdata_230711



testdata_200802



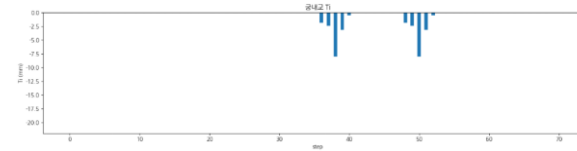
testdata_220712



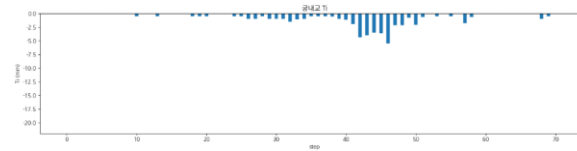
testdata_230711

누적 강우 110mm 이상 4 예측 코드 - **궁내교 2**

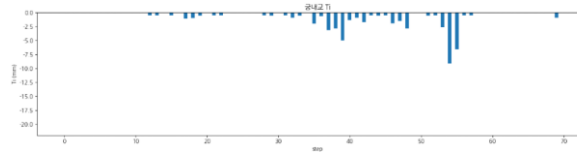
	궁내교
testdata_240723	r2 : -0.0894 mse : 0.1015 mae : 0.2527 rmse : 0.3186
testdata_250716	r2 : 0.8108 mse : 0.0189 mae : 0.1236 rmse : 0.1377
testdata_250814	r2 : 0.8475 mse : 0.0107 mae : 0.0815 rmse : 0.1036



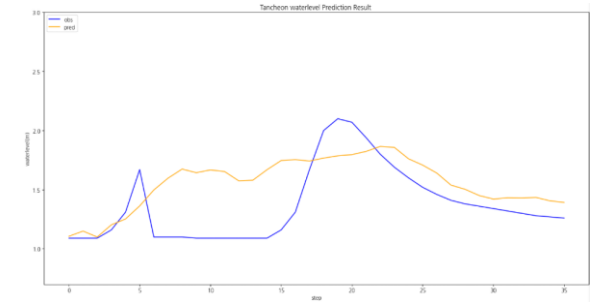
testdata_240723



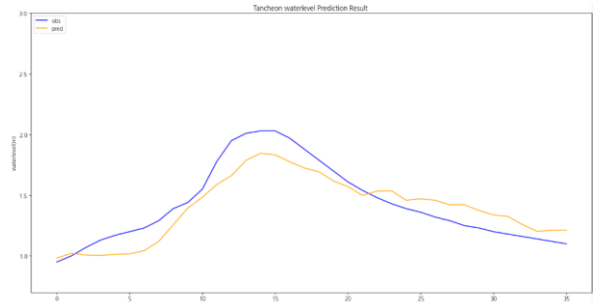
testdata_250716



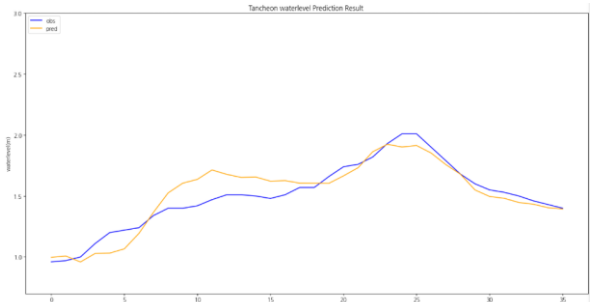
testdata_250814



testdata_240723



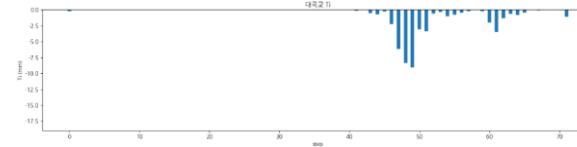
testdata_250716



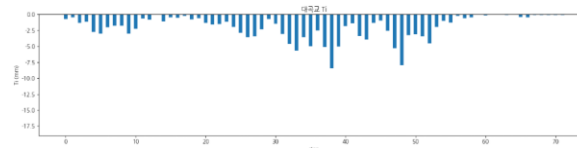
testdata_250814

누적 강우 110mm 이상 4 예측 코드 - 대곡교 1

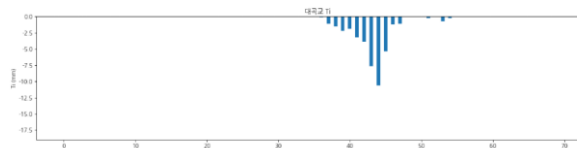
	대곡교
testdata_200802	r2 : 0.8350 mse : 0.1006 mae : 0.2683 rmse : 0.3172
testdata_220712	r2 : -0.0424 mse : 0.1459 mae : 0.3414 rmse : 0.3820
testdata_230711	r2 : 0.7456 mse : 0.2314 mae : 0.3847 rmse : 0.4811



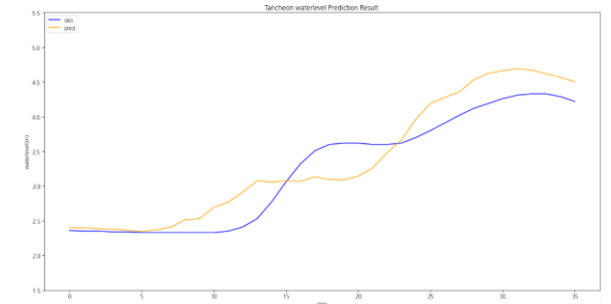
testdata_200802



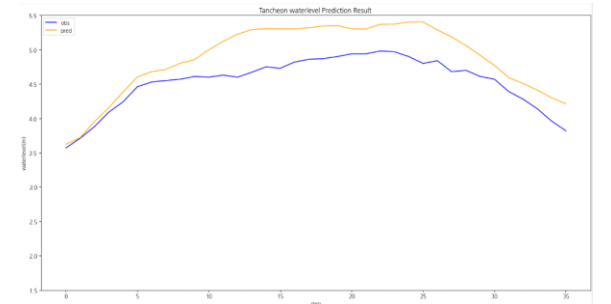
testdata_220712



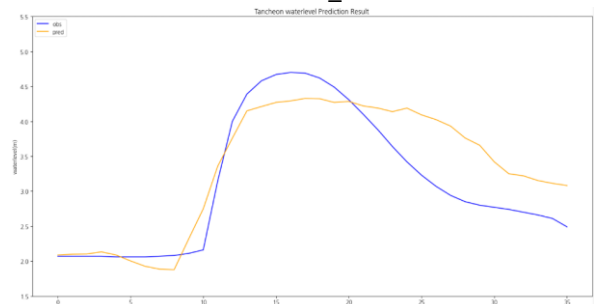
testdata_230711



testdata_200802



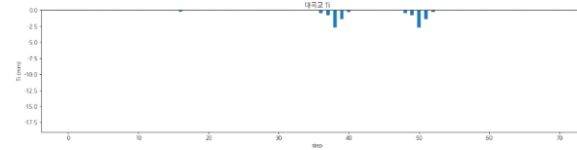
testdata_220712



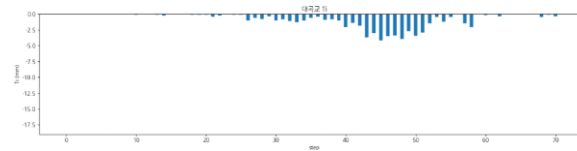
testdata_230711

누적 강우 110mm 이상 4 예측 코드 - 대곡교 2

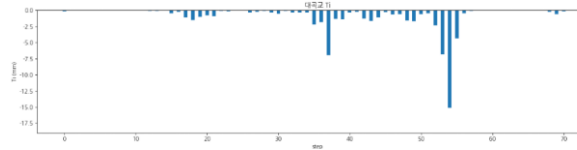
	대곡교
testdata_240723	r2 : -0.7705 mse : 0.0945 mae : 0.2284 rmse : 0.3075
testdata_250716	r2 : 0.8305 mse : 0.0782 mae : 0.1994 rmse : 0.2797
testdata_250814	r2 : 0.5011 mse : 0.1731 mae : 0.3295 rmse : 0.4160



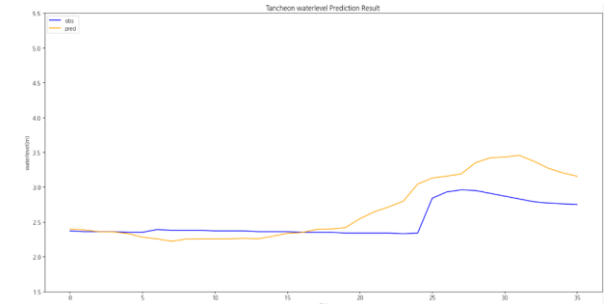
testdata_240723



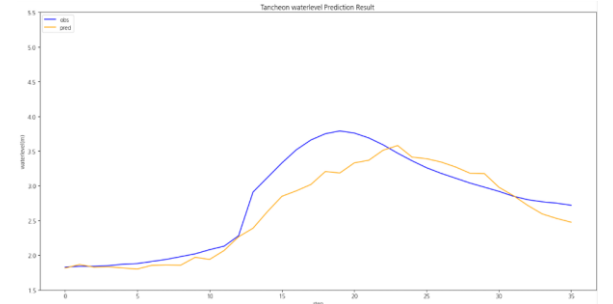
testdata_250716



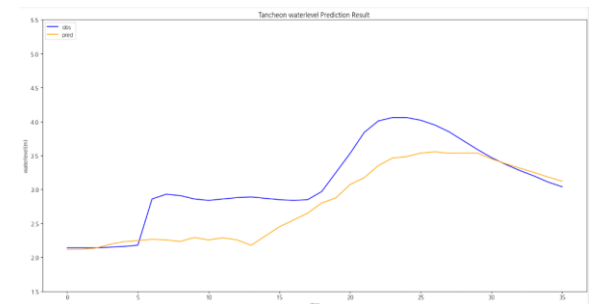
testdata_250814



testdata_240723



testdata_250716



testdata_250814

누적 강우 110mm 이상 5

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0001
 - Loss_function = MSE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

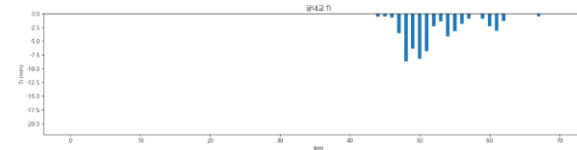
학습 : 총 11개 中 9개 데이터 파일 학습에 사용 (검증 2개)

검증 : 학습 데이터에서 제외한 2개 + 4개 이벤트 사용

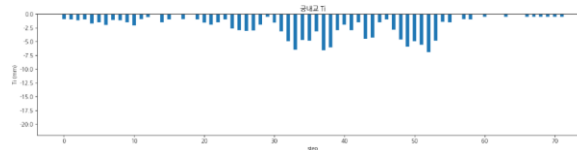
- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022(35) (1개)
- 테스트 : 2022(6667) (1개)

누적 강우 110mm 이상 5 예측 코드 - **궁내교 1**

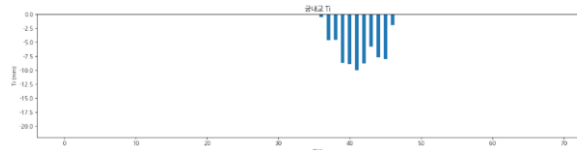
	궁내교
testdata_200802	r2 : 0.8774 mse : 0.0259 mae : 0.1047 rmse : 0.1609
testdata_220712	r2 : 0.0586 mse : 0.0373 mae : 0.1603 rmse : 0.1932
testdata_230711	r2 : 0.8028 mse : 0.0851 mae : 0.1455 rmse : 0.2349



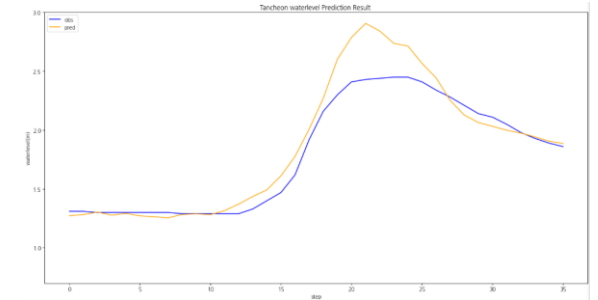
testdata_200802



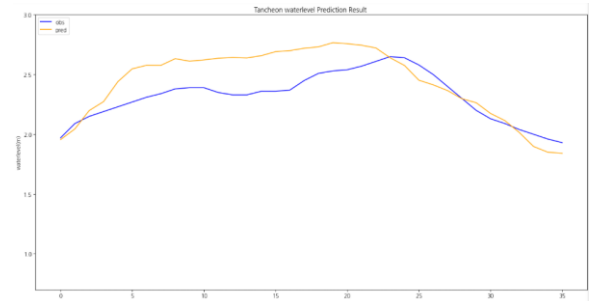
testdata_220712



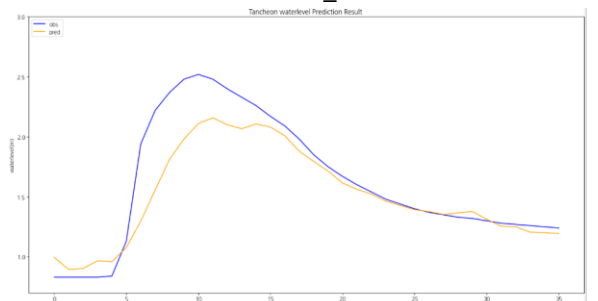
testdata_230711



testdata_200802



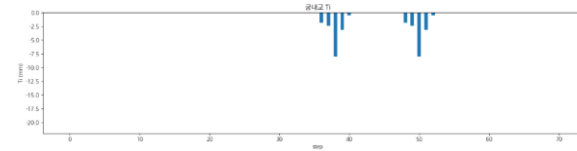
testdata_220712



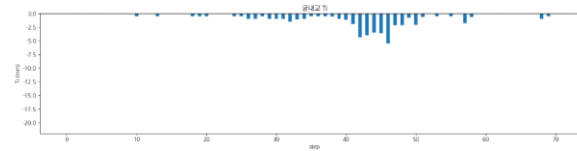
testdata_230711

누적 강우 110mm 이상 5 예측 코드 - **궁내교 2**

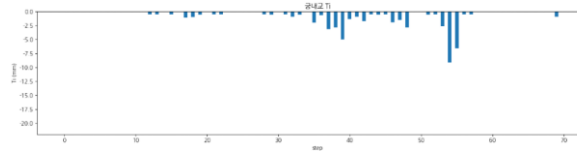
	궁내교
testdata_240723	r2 : 0.1532 mse : 0.0789 mae : 0.2090 rmse : 0.2809
testdata_250716	r2 : 0.7764 mse : 0.0224 mae : 0.1110 rmse : 0.1497
testdata_250814	r2 : 0.8673 mse : 0.0093 mae : 0.0756 rmse : 0.0967



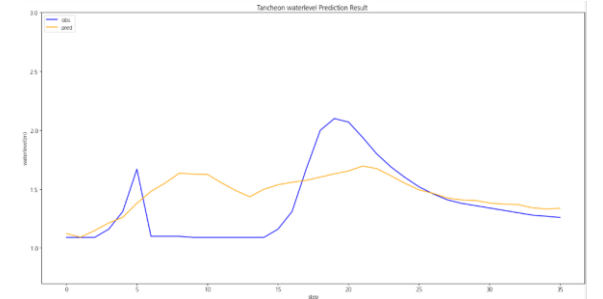
testdata_240723



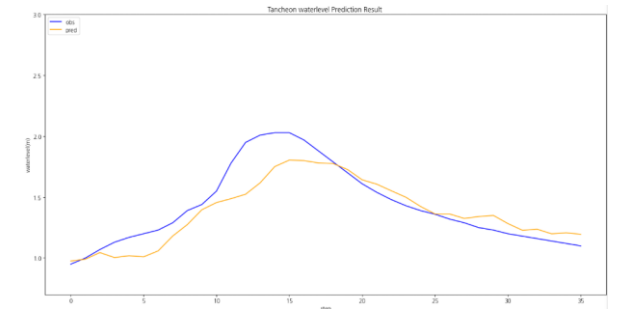
testdata_250716



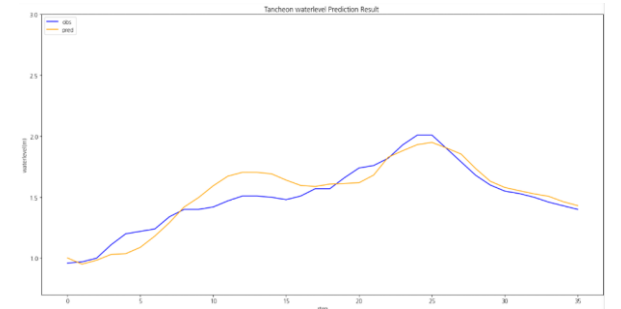
testdata_250814



testdata_240723



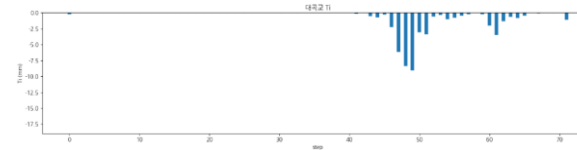
testdata_250716



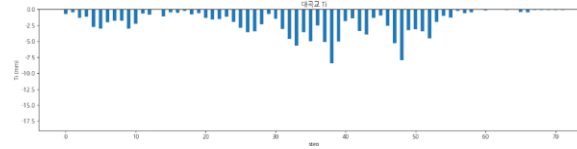
testdata_250814

누적 강우 110mm 이상 5 예측 코드 - 대곡교 1

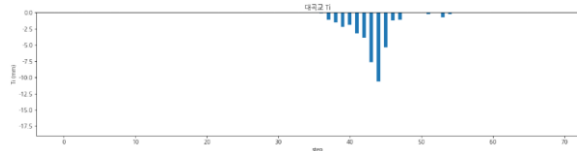
	대곡교
testdata_200802	r2 : 0.9246 mse : 0.0460 mae : 0.1541 rmse : 0.2144
testdata_220712	r2 : -0.1000 mse : 0.1540 mae : 0.3041 rmse : 0.3924
testdata_230711	r2 : 0.8276 mse : 0.1568 mae : 0.3195 rmse : 0.3961



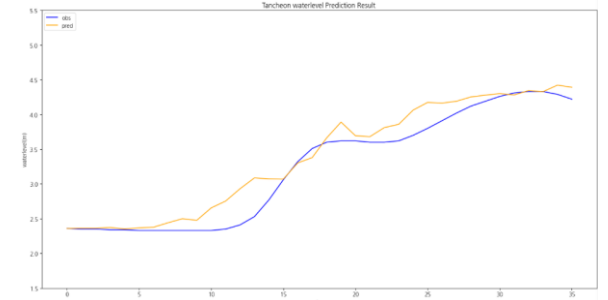
testdata_200802



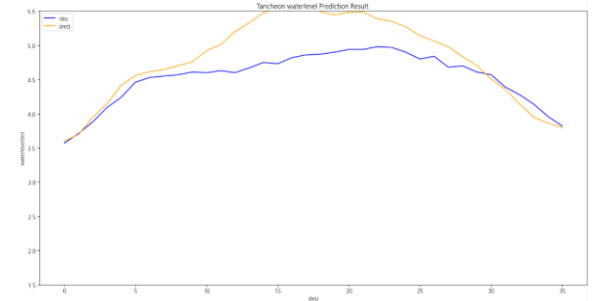
testdata_220712



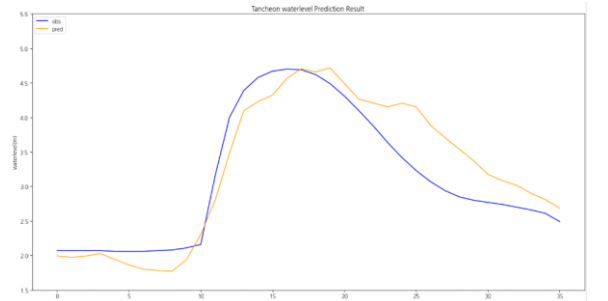
testdata_230711



testdata_200802



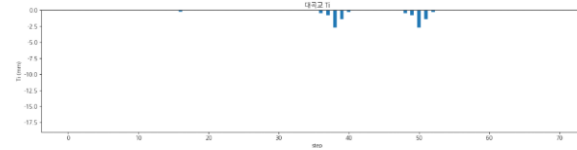
testdata_220712



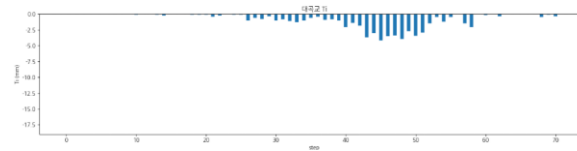
testdata_230711

누적 강우 110mm 이상 5 예측 코드 - 대곡교 2

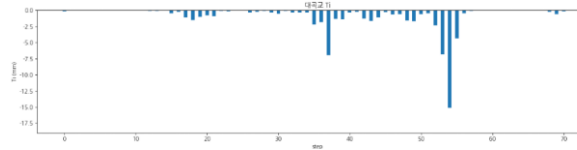
	대곡교
testdata_240723	r2 : -0.551 mse : 0.1258 mae : 0.2542 rmse : 0.3547
testdata_250716	r2 : 0.9577 mse : 0.0195 mae : 0.1145 rmse : 0.1397
testdata_250814	r2 : 0.6115 mse : 0.1348 mae : 0.3007 rmse : 0.3671



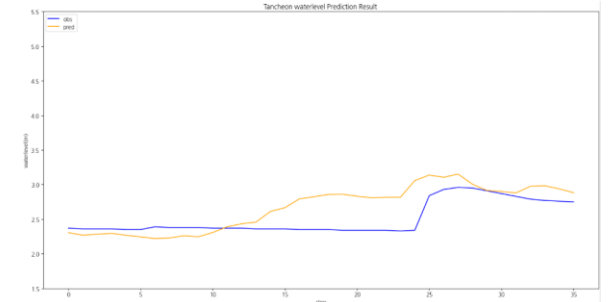
testdata_240723



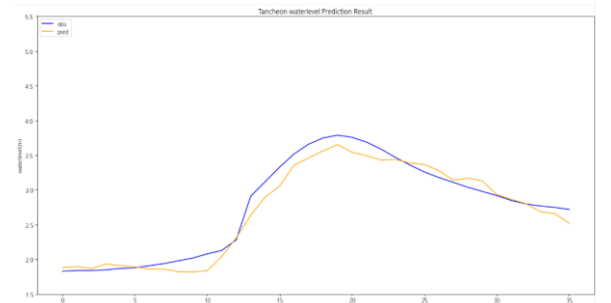
testdata_250716



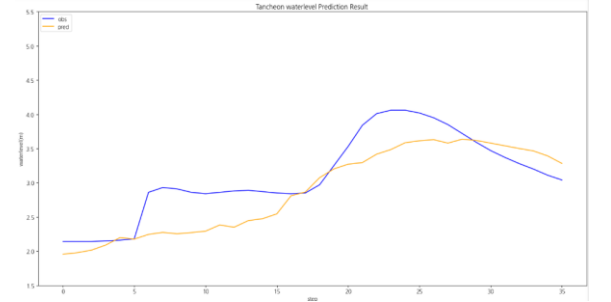
testdata_250814



testdata_240723



testdata_250716



testdata_250814

관심 수위 이상
MAE + 0.0005

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

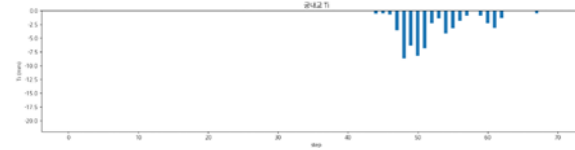
학습 : 총 67개 中 61개 데이터 파일 학습에 사용 (검증 6개)

검증 : 학습 데이터에서 제외한 6개 + 학습에서 검증에 사용된 2개 이벤트 사용

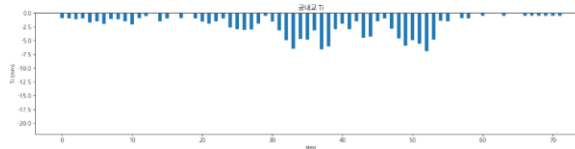
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 이상 + MAE + 0.0005
 예측 코드 - **궁내교 1**

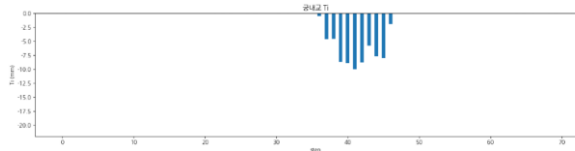
	궁내교
testdata_200802	r2 : 0.9656 mse : 0.0072 mae : 0.0558 rmse : 0.0852
testdata_220712	r2 : 0.4201 mse : 0.0230 mae : 0.1139 rmse : 0.1516
testdata_230711	r2 : 0.8622 mse : 0.0385 mae : 0.1486 rmse : 0.1963



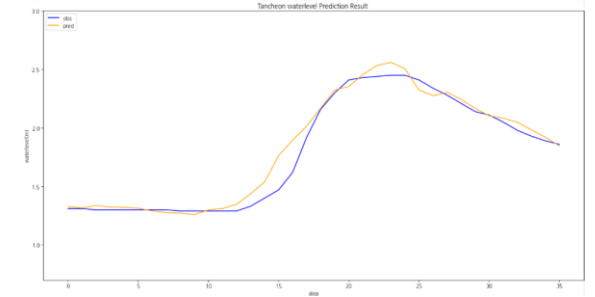
testdata_200802



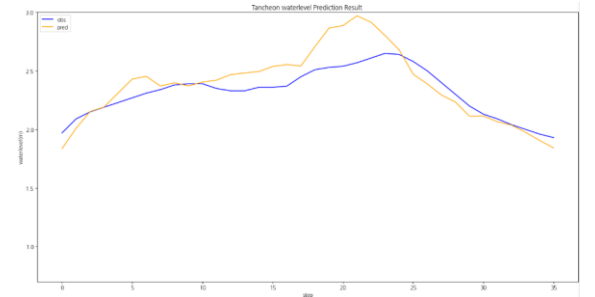
testdata_220712



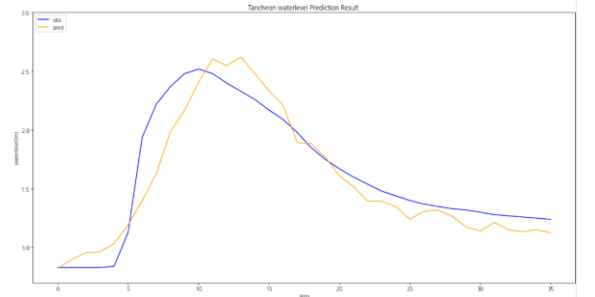
testdata_230711



testdata_200802



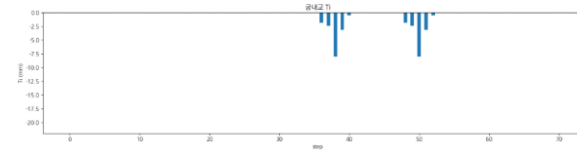
testdata_220712



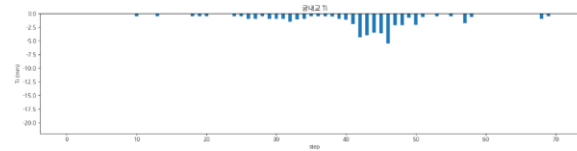
testdata_230711

관심 수위 이상 + MAE + 0.0005
 예측 코드 - **궁내교 2**

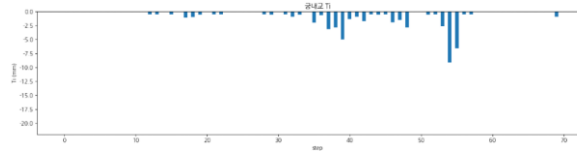
	궁내교
testdata_240723	r2 : -0.0236 mse : 0.0953 mae : 0.2271 rmse : 0.3088
testdata_250716	r2 : 0.7947 mse : 0.0205 mae : 0.1047 rmse : 0.1435
testdata_250814	r2 : 0.5087 mse : 0.0346 mae : 0.1446 rmse : 0.1861



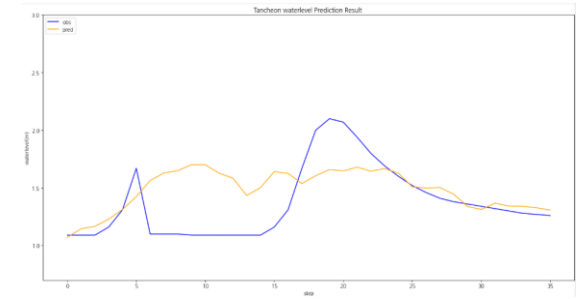
testdata_240723



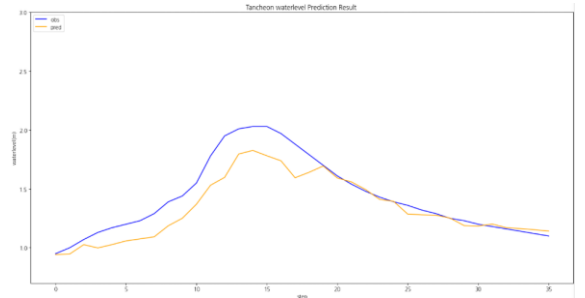
testdata_250716



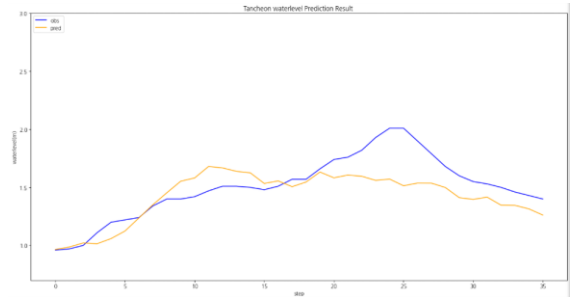
testdata_250814



testdata_240723



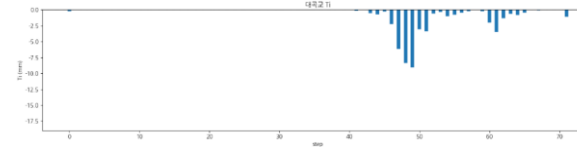
testdata_250716



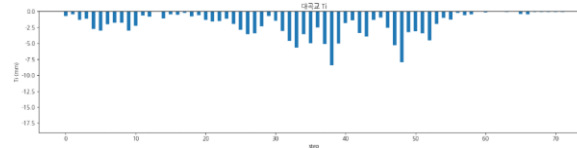
testdata_250814

관심 수위 이상 + MAE + 0.0005 예측 코드 - 대곡교 1

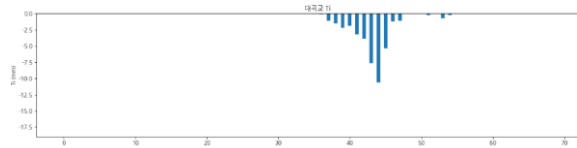
	대곡교
testdata_200802	r2 : 0.9086 mse : 0.0557 mae : 0.1438 rmse : 0.2361
testdata_220712	r2 : 0.8939 mse : 0.0148 mae : 0.0947 rmse : 0.1218
testdata_230711	r2 : 0.9252 mse : 0.0680 mae : 0.2035 rmse : 0.2608



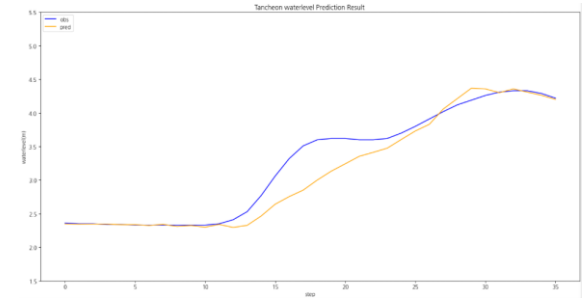
testdata_200802



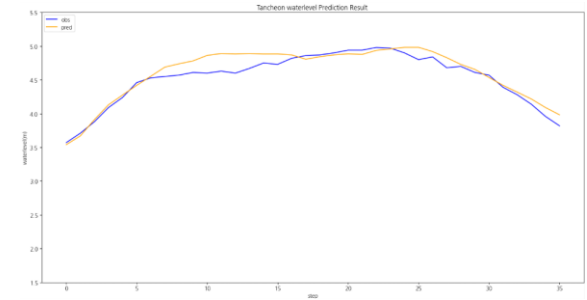
testdata_220712



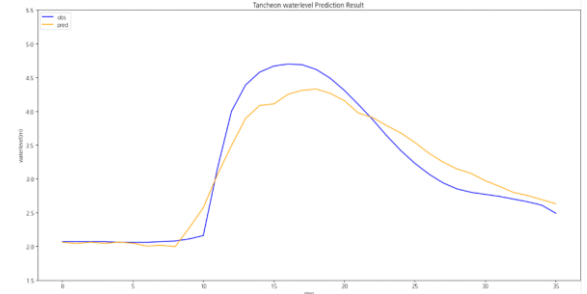
testdata_230711



testdata_200802



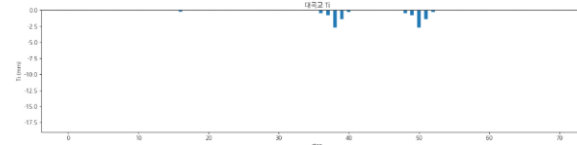
testdata_220712



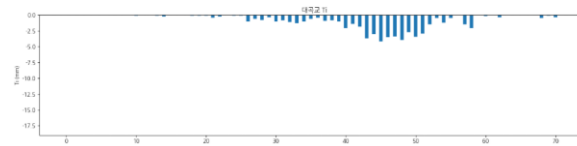
testdata_230711

관심 수위 이상 + MAE + 0.0005 예측 코드 - 대곡교 2

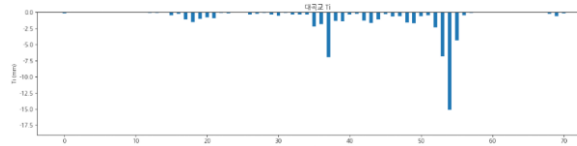
	대곡교
testdata_240723	r2 : -2.7619 mse : 0.2009 mae : 0.3459 rmse : 0.4482
testdata_250716	r2 : 0.9579 mse : 0.0194 mae : 0.0938 rmse : 0.1393
testdata_250814	r2 : 0.7833 mse : 0.0752 mae : 0.2001 rmse : 0.2742



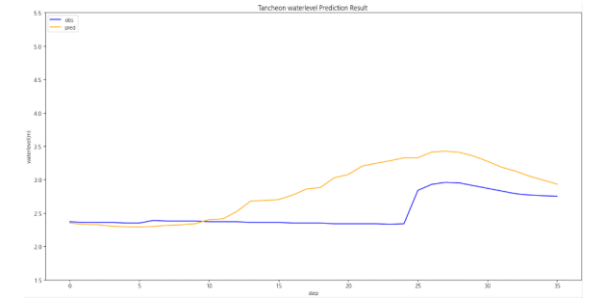
testdata_240723



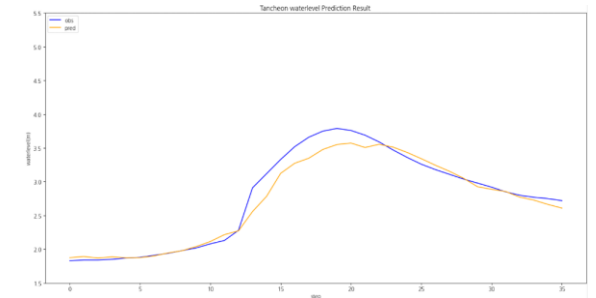
testdata_250716



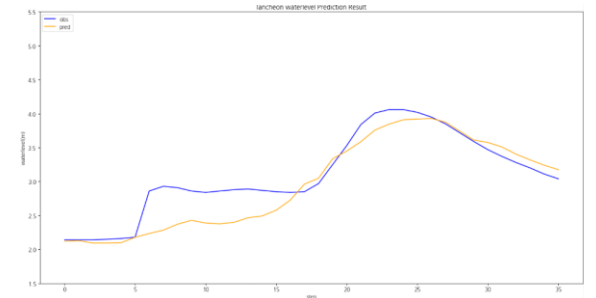
testdata_250814



testdata_240723



testdata_250716



testdata_250814

관심 수위 이상 huber

작은 오차는 MSE처럼 부드럽게,

큰 오차(이상치)는 MAE처럼 강건하게

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0001
 - Loss_function = Huber
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

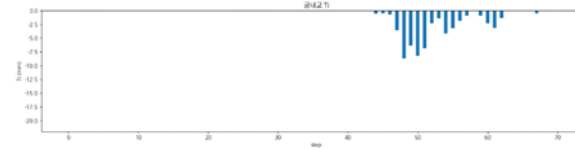
학습 : 총 67개 中 61개 데이터 파일 학습에 사용 (검증 6개)

검증 : 학습 데이터에서 제외한 6개 + 학습에서 검증에 사용된 2개 이벤트 사용

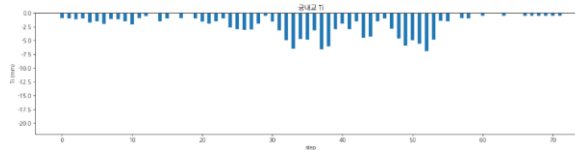
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 이상 + Huber 예측 코드 - **궁내교 1**

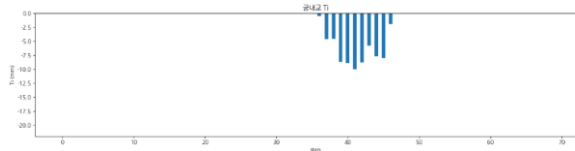
	궁내교
testdata_200802	r2 : 0.9436 mse : 0.0119 mae : 0.0800 rmse : 0.1091
testdata_220712	r2 : 0.5280 mse : 0.0187 mae : 0.1057 rmse : 0.1368
testdata_230711	r2 : 0.8739 mse : 0.0352 mae : 0.1445 rmse : 0.1878



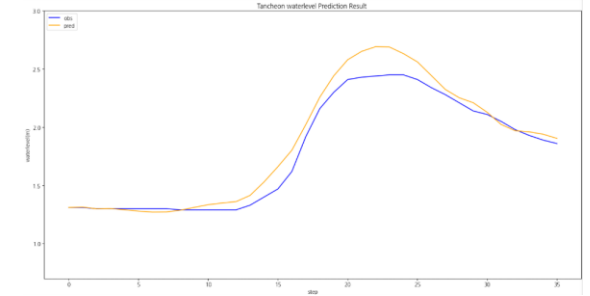
testdata_200802



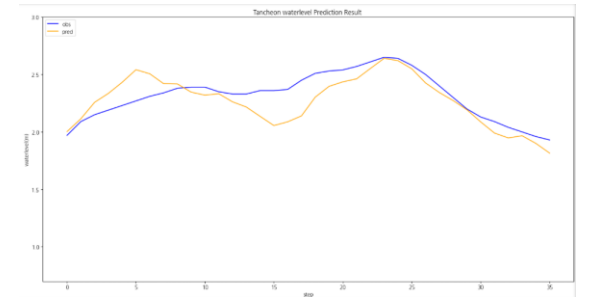
testdata_220712



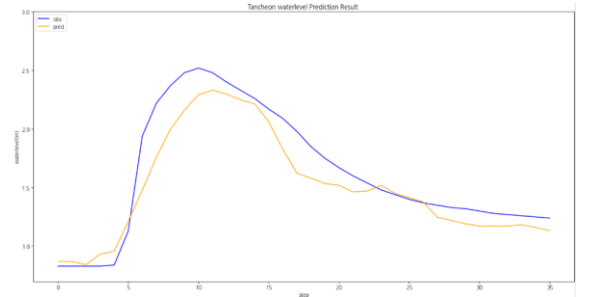
testdata_230711



testdata_200802



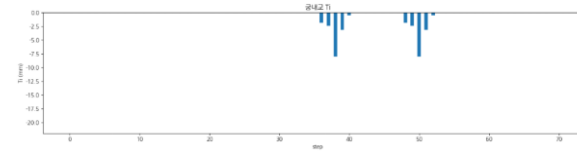
testdata_220712



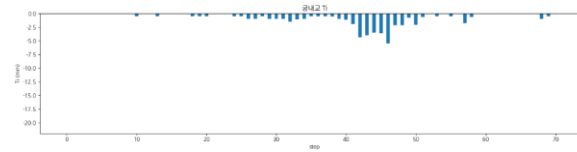
testdata_230711

관심 수위 이상 + Huber 예측 코드 - 궁내교 2

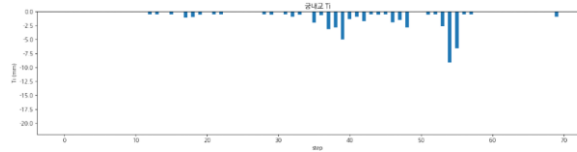
	궁내교
testdata_240723	r2 : 0.0597 mse : 0.0876 mae : 0.2157 rmse : 0.2960
testdata_250716	r2 : 0.8570 mse : 0.0143 mae : 0.0951 rmse : 0.1197
testdata_250814	r2 : 0.6947 mse : 0.0215 mae : 0.1217 rmse : 0.1467



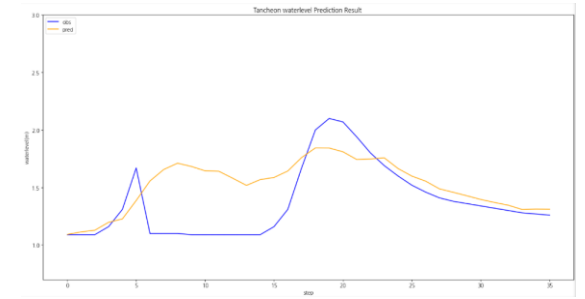
testdata_240723



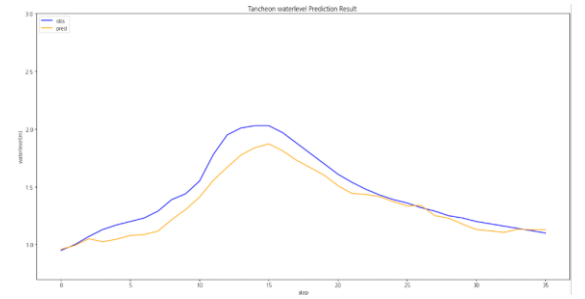
testdata_250716



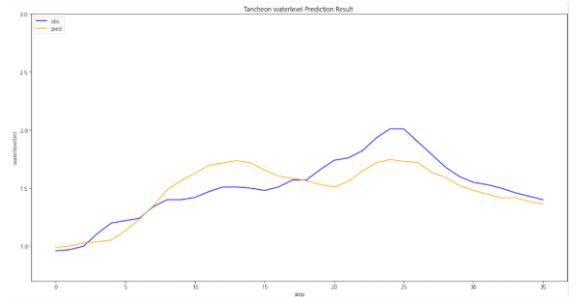
testdata_250814



testdata_240723



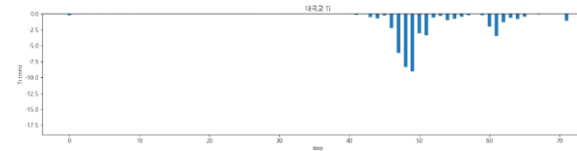
testdata_250716



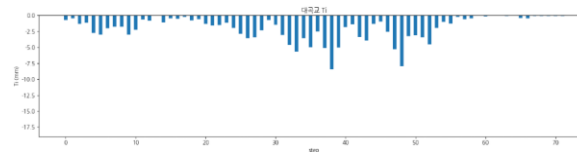
testdata_250814

관심 수위 이상 + Huber 예측 코드 - 대곡교 1

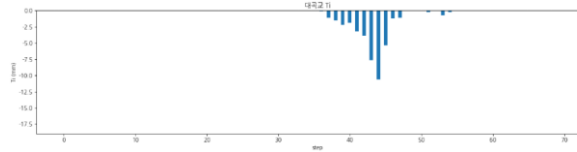
	대곡교
testdata_200802	r2 : 0.8583 mse : 0.0864 mae : 0.1800 rmse : 0.2939
testdata_220712	r2 : 0.9114 mse : 0.0123 mae : 0.0931 rmse : 0.1113
testdata_230711	r2 : 0.8354 mse : 0.1497 mae : 0.2952 rmse : 0.3870



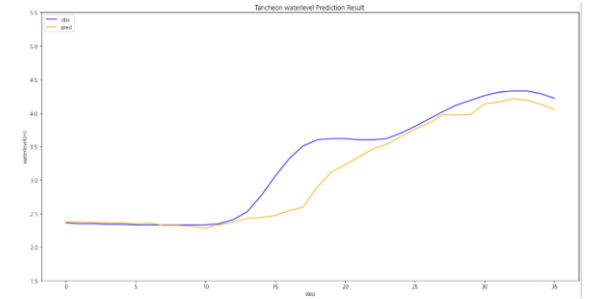
testdata_200802



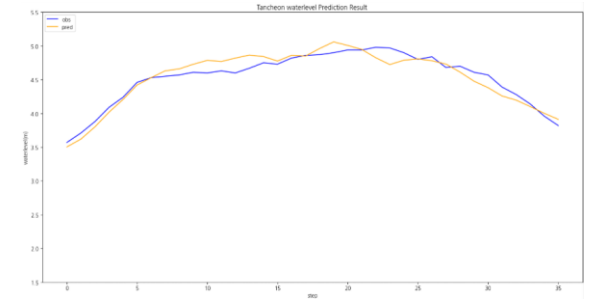
testdata_220712



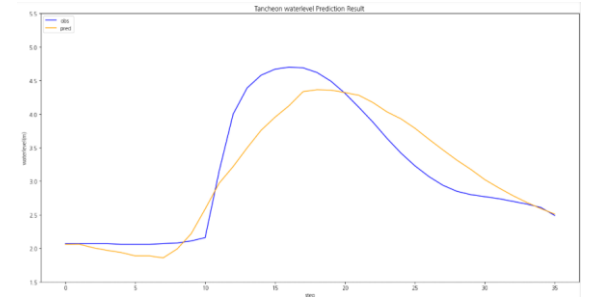
testdata_230711



testdata_200802



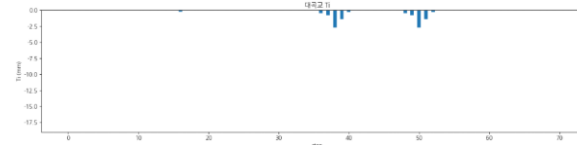
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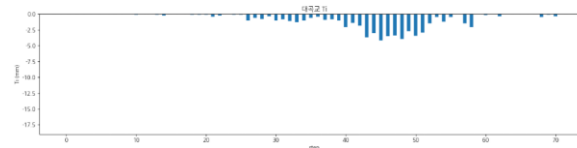
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관심 수위 이상 + Huber 예측 코드 - 대곡교 2

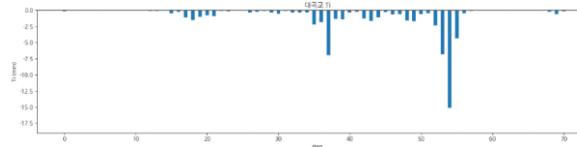
	대곡교
testdata_240723	r2 : -0.3854 mse : 0.0739 mae : 0.1788 rmse : 0.2720
testdata_250716	r2 : 0.9566 mse : 0.0200 mae : 0.1153 rmse : 0.1414
testdata_250814	r2 : 0.7191 mse : 0.0974 mae : 0.2343 rmse : 0.3122



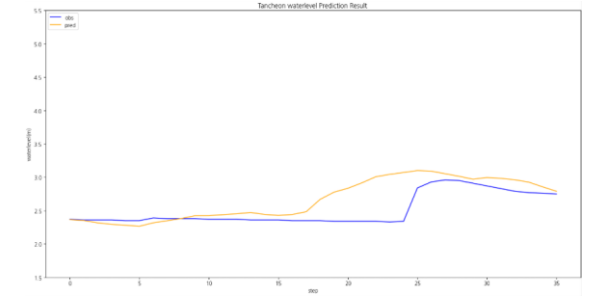
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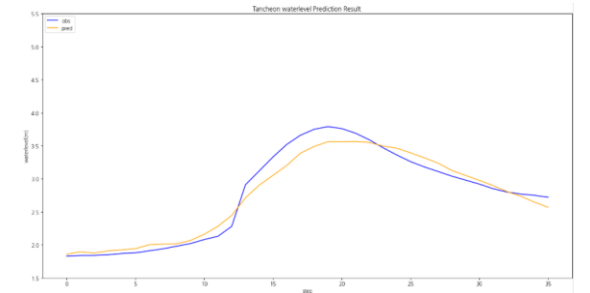
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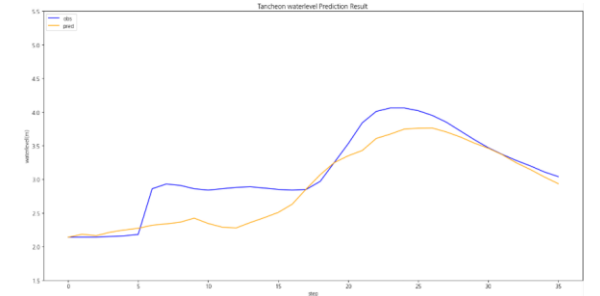
testdata_250814



testdata_240723



testdata_250716



testdata_250814